The Japan Society of Mechanical Engineers
No.15-210 Proceedings of the 6th International
Conference on Advanced Mechatronics (ICAM2015),
Tokyo, Japan, December 5-8, 2015

Evaluation of Breaking Performance in Vibration-Assisted Electrostatic Surface Induction Actuator

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This paper evaluates breaking performance of an electrostatic surface induction actuator. The actuator is equipped with piezoelectric vibrators such that the friction between the slider and the stator electrodes can be dramatically reduced by squeeze-film effect. In such an actuator, the friction force can be changed by turning on and off the vibrator. The friction change can be utilized for high-performance slider motion control; for example, friction can be increased by switching off the vibrator when the slider needs to stop. In this paper, we evaluated how fast the slider can stop in several conditions. The result clearly shows the effect of friction change in breaking performance of the actuator.

1 Introduction

Table top displays with electrostatic surface actuators enable users to interact with computers through real physical objects placed on the display [1, 2]. The electrostatic surface actuators can be fabricated thin and transparent by utilizing screen printing; printing fine-pitch electrodes using transparent (or translucent) conductive paste on a transparent plastic sheet or glass plate can realize thin and transparent actuator. Such an actuator can be quite easily mounted on off-the-shelf LCD monitors. For example, in [1], electrostatic surface induction actuator was placed on LCD monitor to realize physical interaction with computer-animated objects; the user can move a paper sheet placed on the LCD to play with a computer-animated object inside the monitor.

One of the technical challenges in the electrostatic surface actuators is the reduction of friction between the slider and the stator electrodes. The electrostatic force that drives the slider in horizontal direction also works in vertical direction as attraction force. The attraction force is much larger than the lateral driving force, and therefore, the interface surface between the slider and the stator must have very low surface friction; otherwise the slider does not move due to large friction force. One common method to reduce friction coefficient is to spread tiny glass beads between slider and stator [1, 2]. However, use of glass beads raises a maintenance problem. Another method is to use special surface coating for both surfaces of the slider and the stator [3]. The surface coating, however, limits the slider material.

Recently, another method of friction reduction has been tried, which utilizes squeeze film effect caused by vibration [4]. By vibrating the stator surface using piezoelectric elements, a thin air layer is produced between the slider and the stator, which dramatically reduces the friction.

The use of squeeze film effect brings us a new function that does not exist in the previous means of friction reduction. The new function is modulation of friction force; by turning on and off the piezoelectric vibrator, friction coefficient can virtually be changed. For example, friction can be reduced for acceleration of the slider, whereas friction can be enhanced for sudden stop of the slider. Such characteristic is quite important for the above-mentioned application of the electrostatic actuator. In the interactive system reported in [1], the slider needs to stop instantly when it hits the virtual object rendered in the LCD.

Toward such goal, this paper investigates the breaking performance of the electrostatic actuator with squeeze-film effect. The performance was compared in three different conditions, where either or both of vibration and electrostatic driving force are turned off to stop the slider.

2 Experiment

The experimental setup is shown in Fig. 1. The electrostatic actuator was made by screen printing. Translucent conductive paste was printed on a glass plate to form fine-pitch electrodes. The electrode pattern consists of 16 sub-areas arranged in 4-by-4 matrix, as same as in [2]. Each sub-area contains three-phase electrode arrangements in x- or y-direction. Such electrode pattern can realize rotation of the slider at some fixed points, in addition to 2-DOF translational motion [2]. On both edges of the glass plate, vibration sources are attached, consisting of beryllium copper plates and piezoelectric elements. The detail of the actuator design is reported in [4].

The breaking performance was evaluated for rotation motion of the slider, as there is no stroke limitation for rotational motions. Transparent slider (OHP sheet) was cut into circular form with a diameter of 80 mm and placed on the stator surface such that the slider covers four electrode

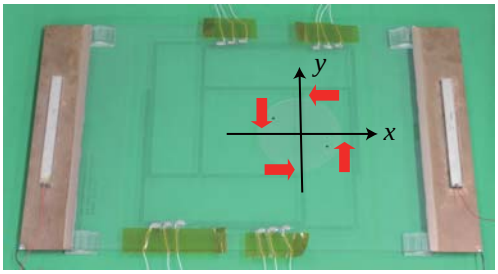


Fig. 1 Experimental setup.

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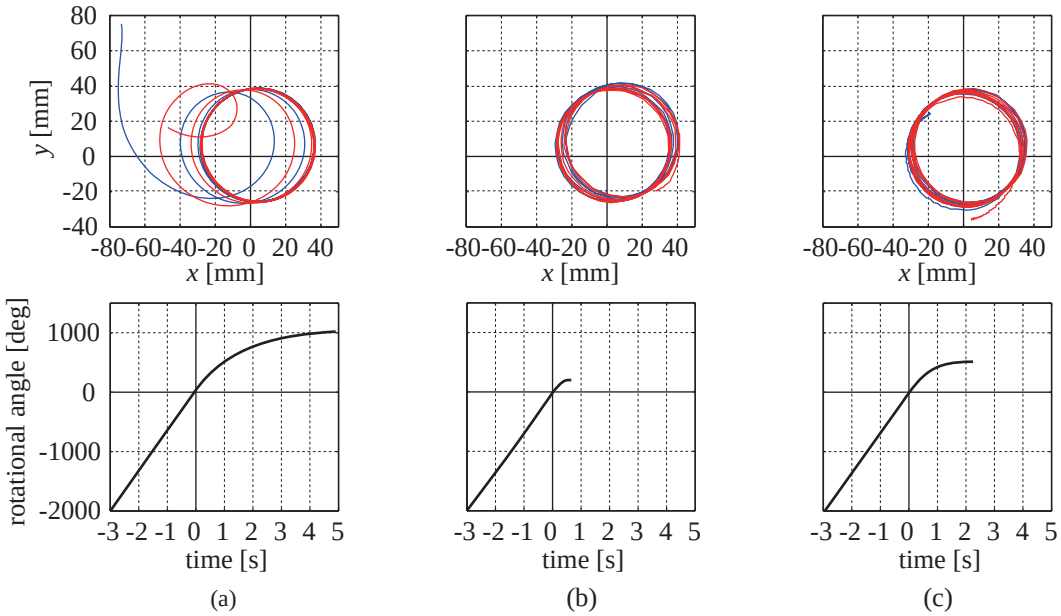


Fig. 2 Trajectories of two fiducial markers and rotational angle of the slider.

sub-areas. The electrodes were fed three-phase voltage such that they generate electrostatic driving force in the directions illustrated in Fig. 1 using red arrows. The motion of the slider was recorded by high speed camera (VW-6000, Keyence). To facilitate it, the slider was equipped with two fiducial markers (black dots in Fig. 1).

The slider was rotationally accelerated using electrostatic driving force until it reaches a steady-state. The vibration was kept on during the acceleration. Then, either electrostatic driving force or vibration, or both, was suddenly turned off to investigate breaking performance. The driving conditions for the piezo elements were 290 $V_{p\text{-}p}$ and 17.9 kHz, and those for electrostatic driving were 500 $V_{p\text{-}p}$ and 200 Hz.

Fig. 2 shows the results. The plots on the top show the motion trajectories of the fiducial markers on the slider, recorded from three seconds before the breaking started until the slider stopped. The plots on the bottom show the time-course of the rotational angle of the slider. The breaking conditions are

(a) only electrostatic driving force was cut off,
(b) only vibration was cut off (electrostatic driving force was kept active), and
(c) both electrostatic driving force and vibration were cut off.

In case (a), the slider lost control and drifted away due to too small friction force. It hit some obstacles, such as electrical terminals, before it completely stopped in about 5 seconds. In case (b), the slider stopped the rotation in about 0.5 second, even though there was still electrostatic driving force. This stopping time was much shorter than the case (c) where the slider needed about two seconds to stop. In case (b), it seems that the electrostatic attraction force became dominant due to the increased friction coefficient after the cut off of the vibration. Comparison of these three cases clearly shows the effect of friction changes for slider breaking performance. This time, in case (b), the voltages for the electrostatic actuator was kept the same even after the cut-off of the vibration. The breaking performance would be much enhanced if we apply different voltage pattern for breaking, such as voltage pattern for reversed motion or DC voltages.

3 Conclusion

Friction reduction between the slider and the stator electrodes was one of the technical challenges in electrostatic surface actuators. Use of squeeze-film effect is one promising mean for friction reduction. In this paper, we focused on another interesting aspect of the squeeze-film effect; the friction between the slider and the stator can be changed. The effect of friction change was evaluated in breaking performance for the rotating slider. The results clearly shows that the breaking performance was enhanced by cutting off the vibration for the squeeze film effect.

Acknowledgement

This work was supported in part by KAKENHI B (No. 26280069) from JSPS, Japan.

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